Covering Tracks

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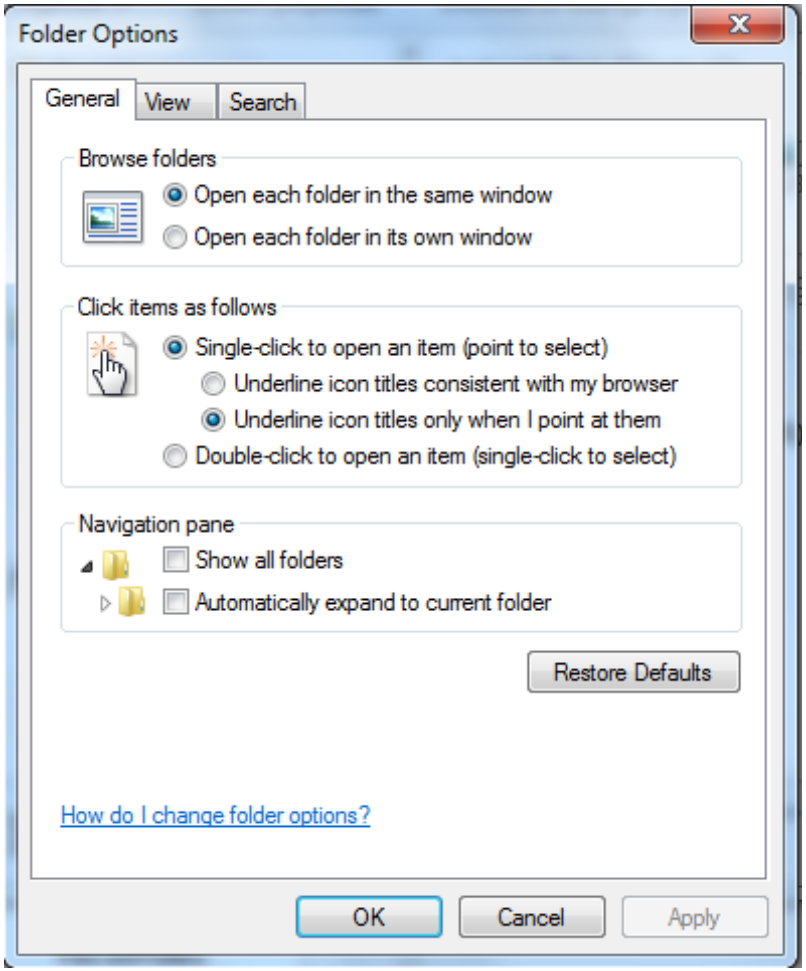
Covering Tracks - Scope

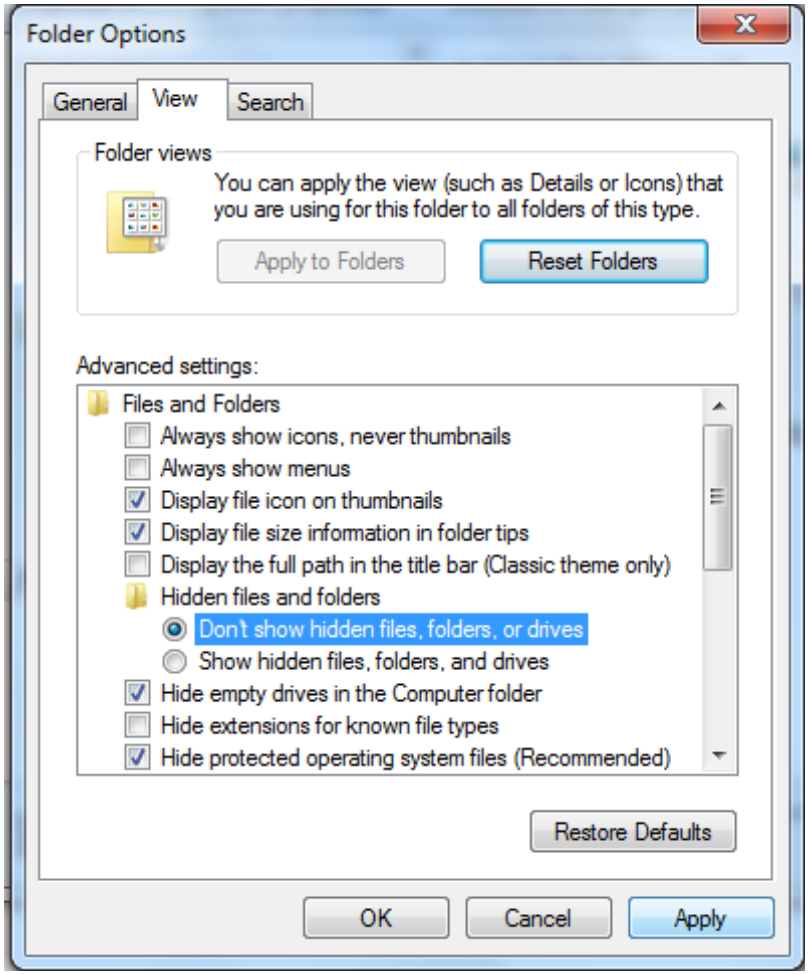
- After an attacker compromises a machine and completes the attack or creates a back door, he has to ensure that his presence is removed or remains hidden:
- Hiding files and folders
- Undeletable files and folders
- Clean up log files
- Clean up target machine



Hiding Files and Folders

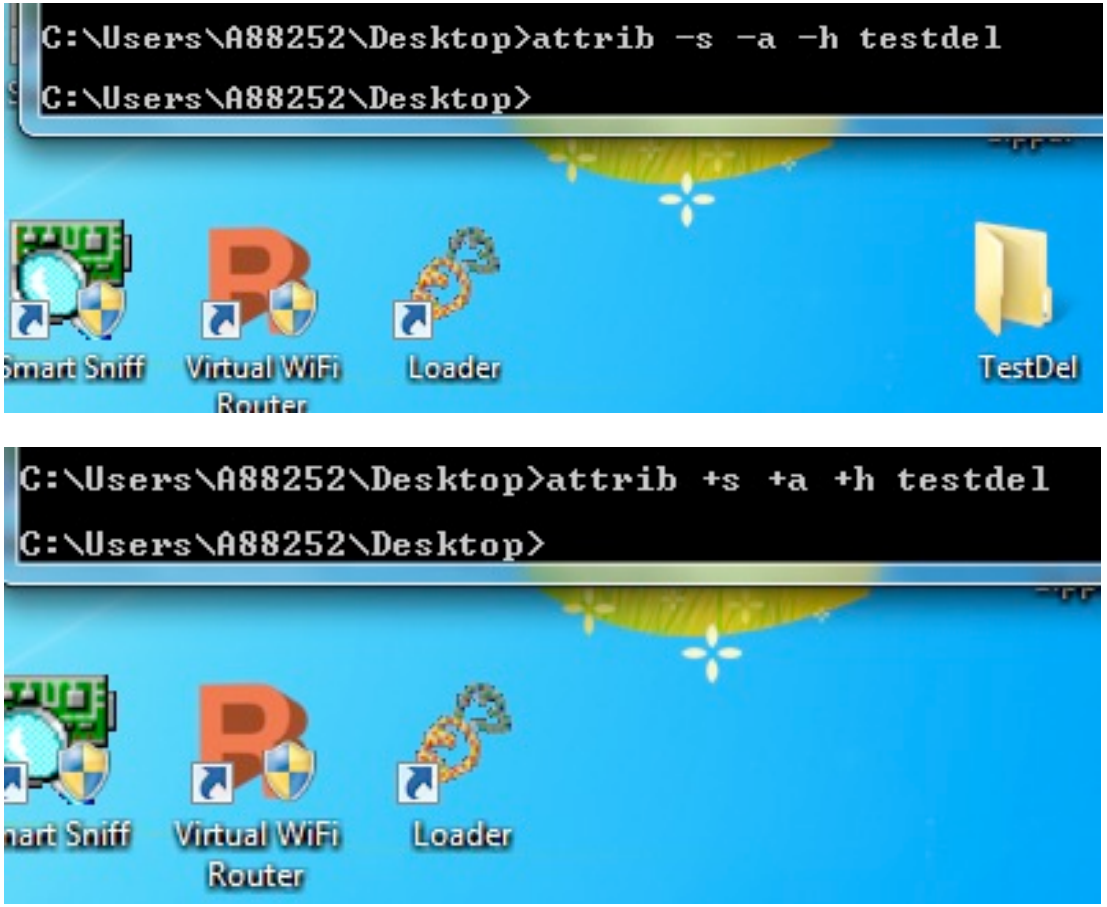
Make It Hidden - Explorer





- Explorer -> Computer -> Organize -> Folder and search optioins
- Folder Options -> View tab -> Don't show hidden …
- Can be applied to files, folders as well as drives

Make It Hidden - Command



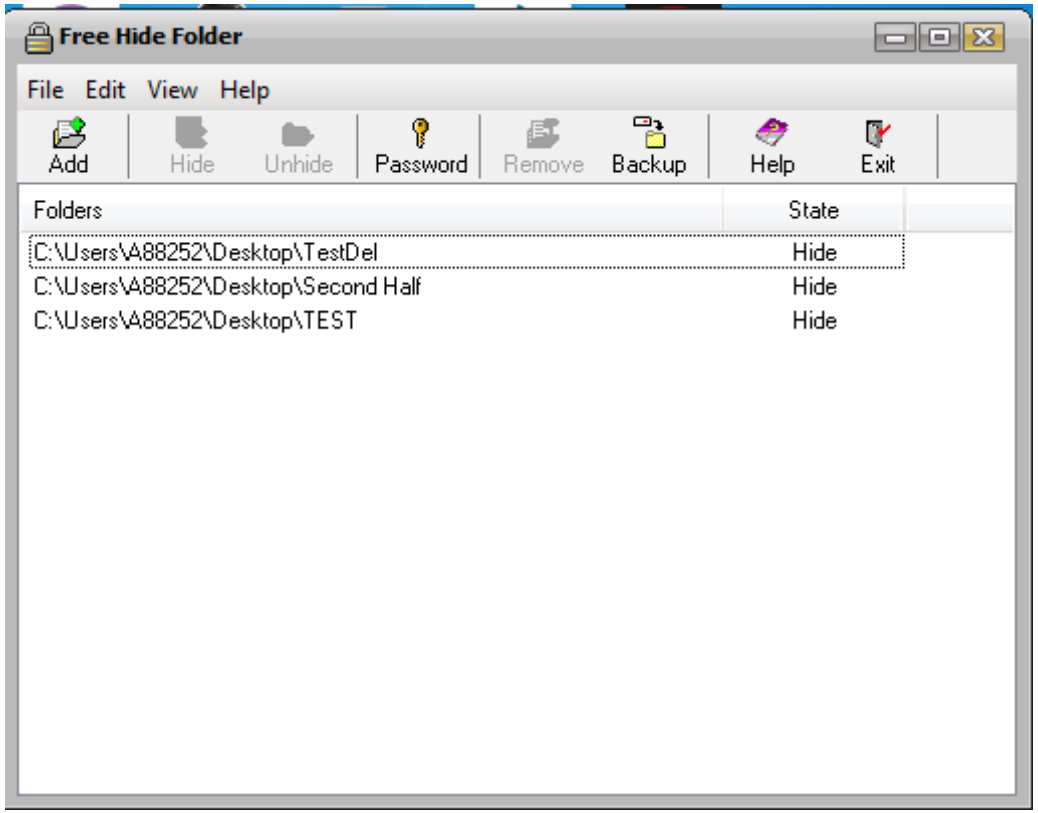
- attrib +s +a +h <foldername>
- Change folder attributes to system, archive and hidden
- Can be applied to files as well

Make It Hidden – Free Hide Folder

SINGAPORE INSTITUTE OF TECHNOLOGY

http://www.cleanersoft.com/hidefolder/free_hide_folder.htm



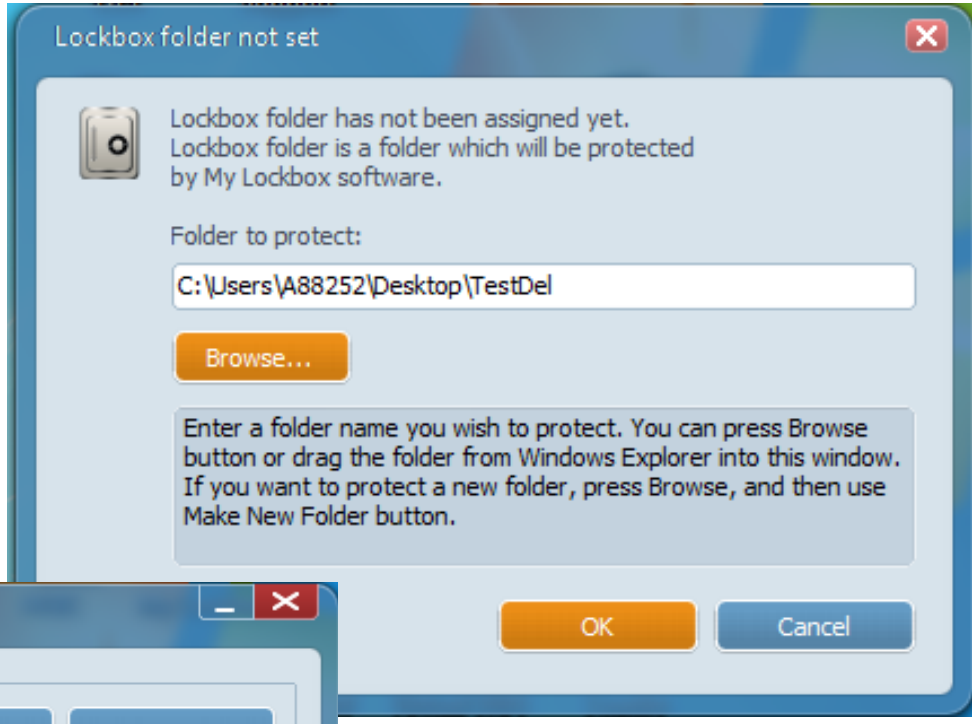


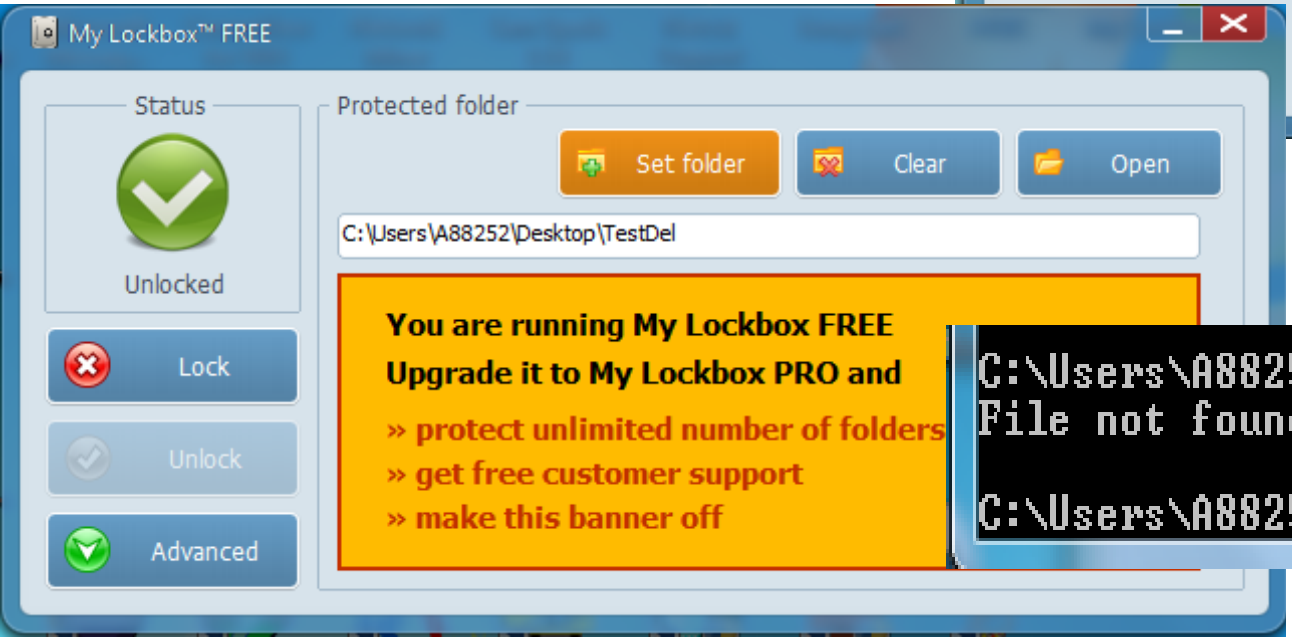
- Click Skip to bypass donation
- Add folders to be hidden (no support for files)
- Can selectively hide and unhide

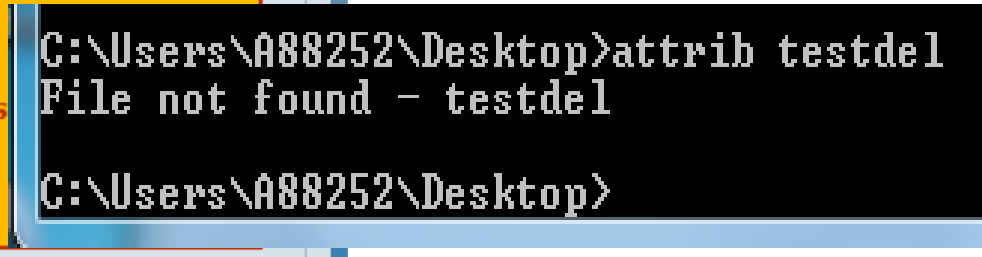
Make It Hidden – My Lockbox

http://fspro.net/my-lockbox/

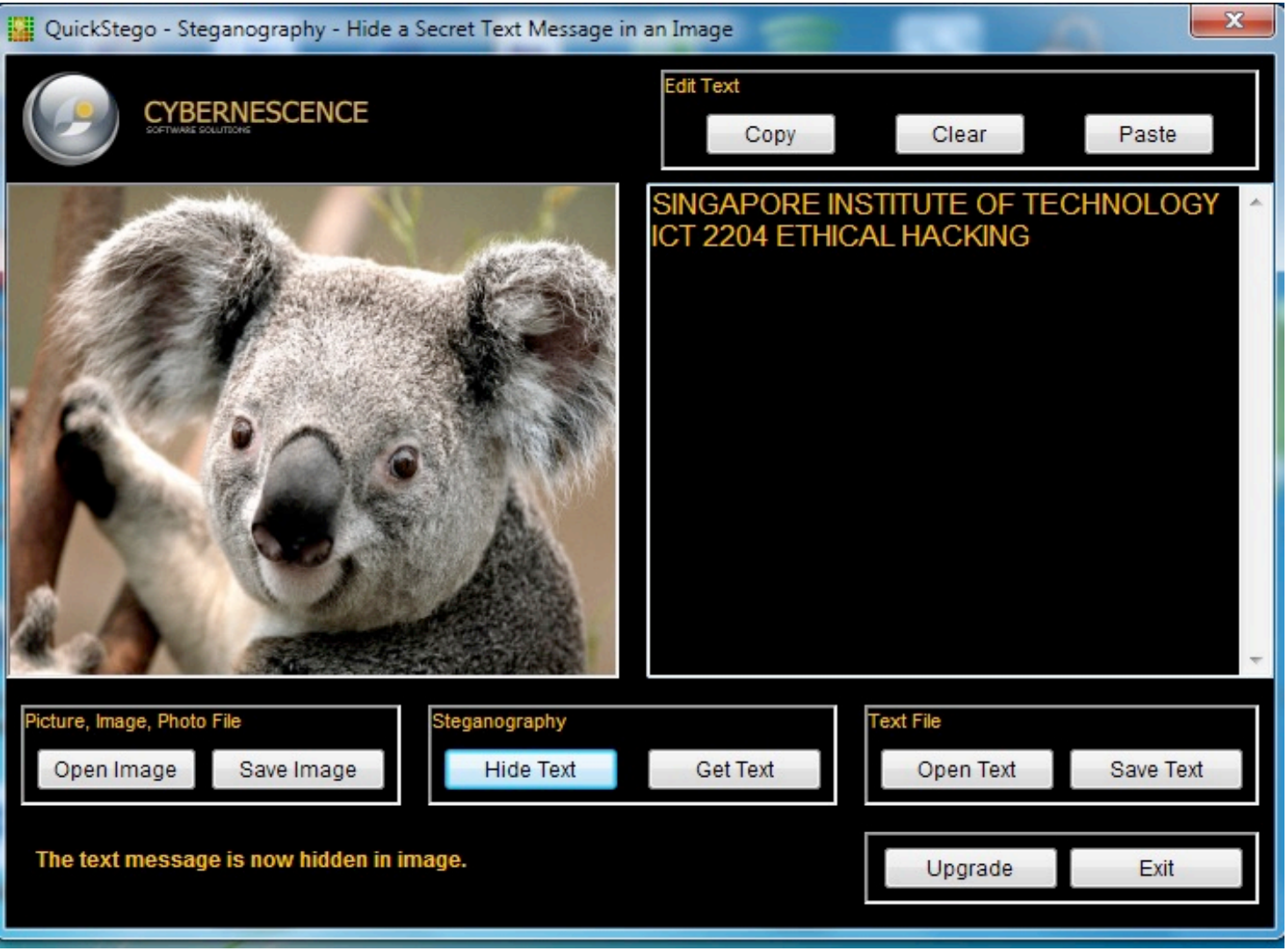
- Add folders to be hidden (no support for files)
- Can selectively lock and unlock







Make It Hidden – QuickStego



http://www.softpedia.com/get/Security/Encrypting/QuickStego.shtml#download

- Hide collection of e.g. usernames, passwords, pass phrases
- Text can be hidden in an innocuous image on the target

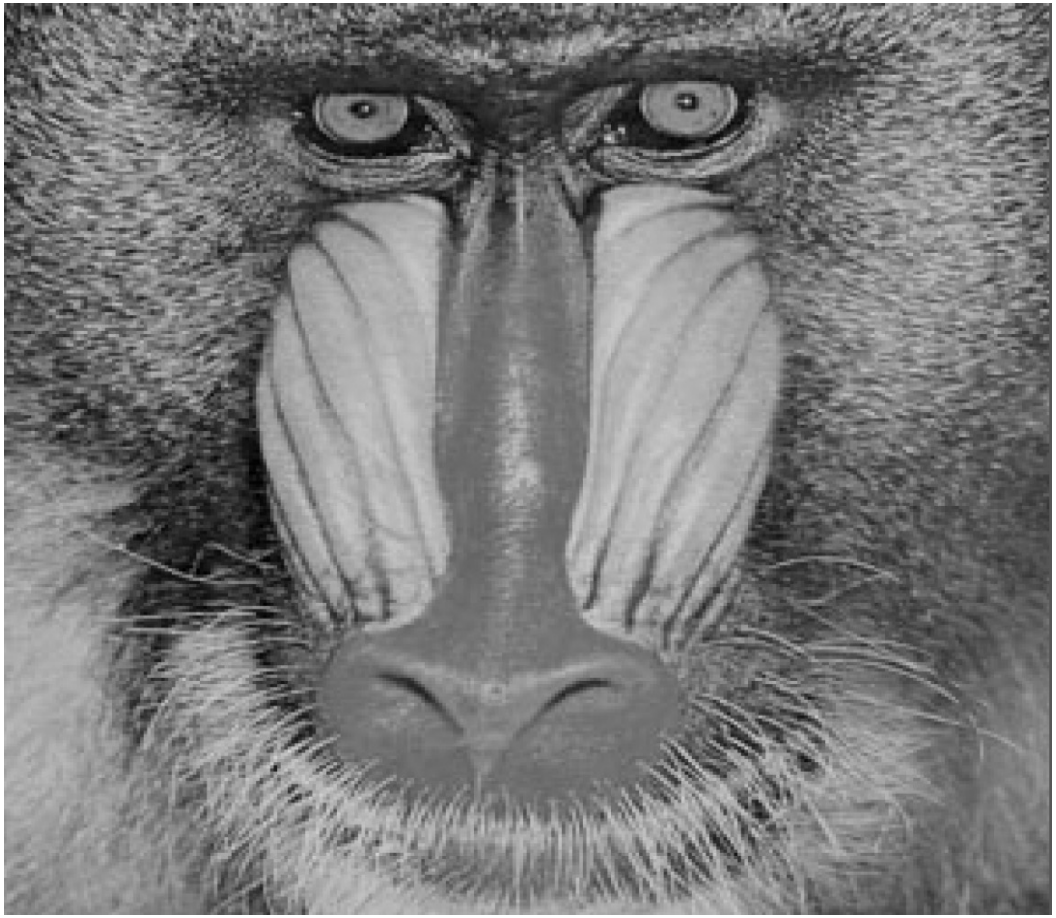
Stego Image File

- Image files contain header information and image pixels
- The header information is usually quite precise, with little spare capacity
- Much of the file comprises image pixels
- Each pixel contain the 3 primary colour components: **red**, **green** and **blue**
- Each colour component is 8 bits long, giving 256 different possible values
- A full colour image contains 3 colour component planes
- A green colour component plane example follows

Full Colour Image

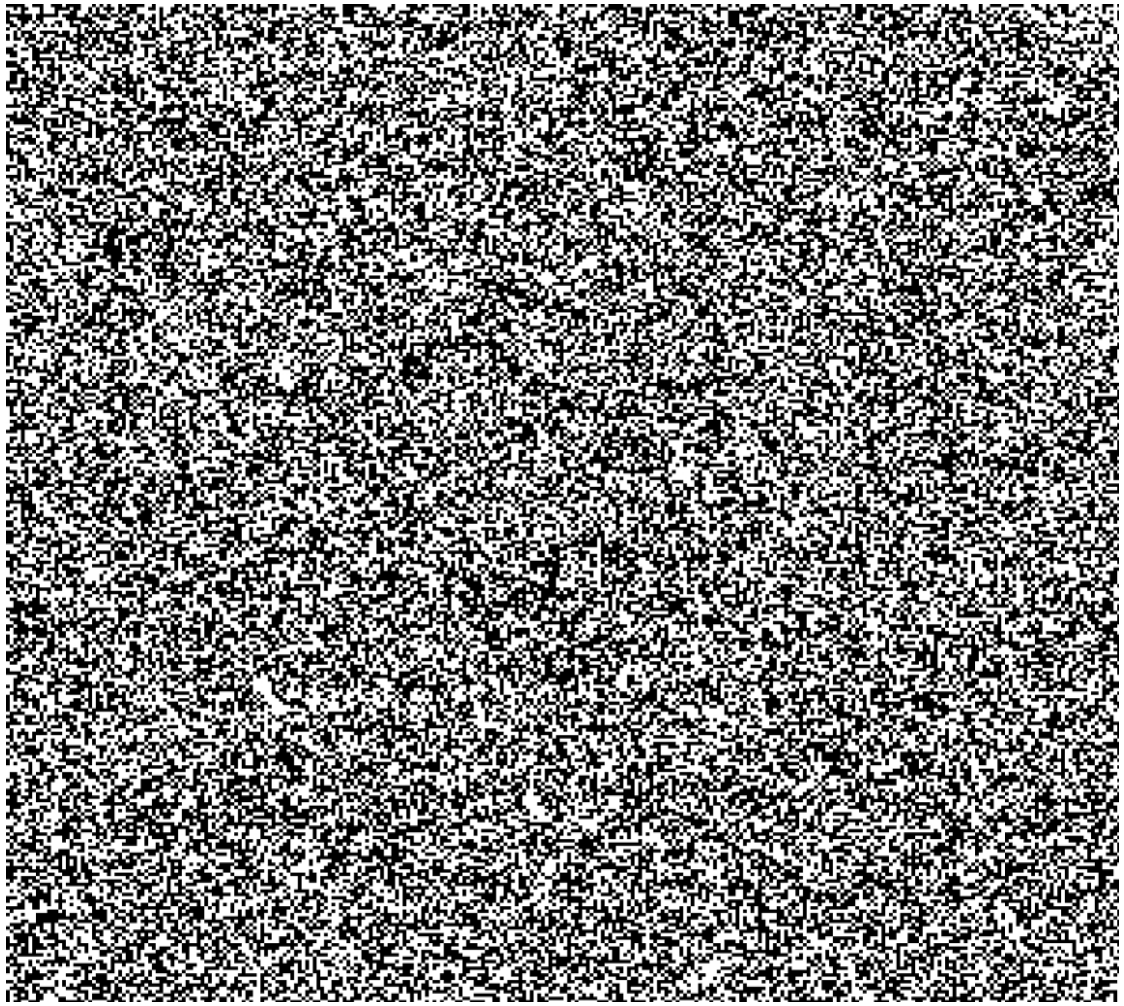


Green Colour Component Plane

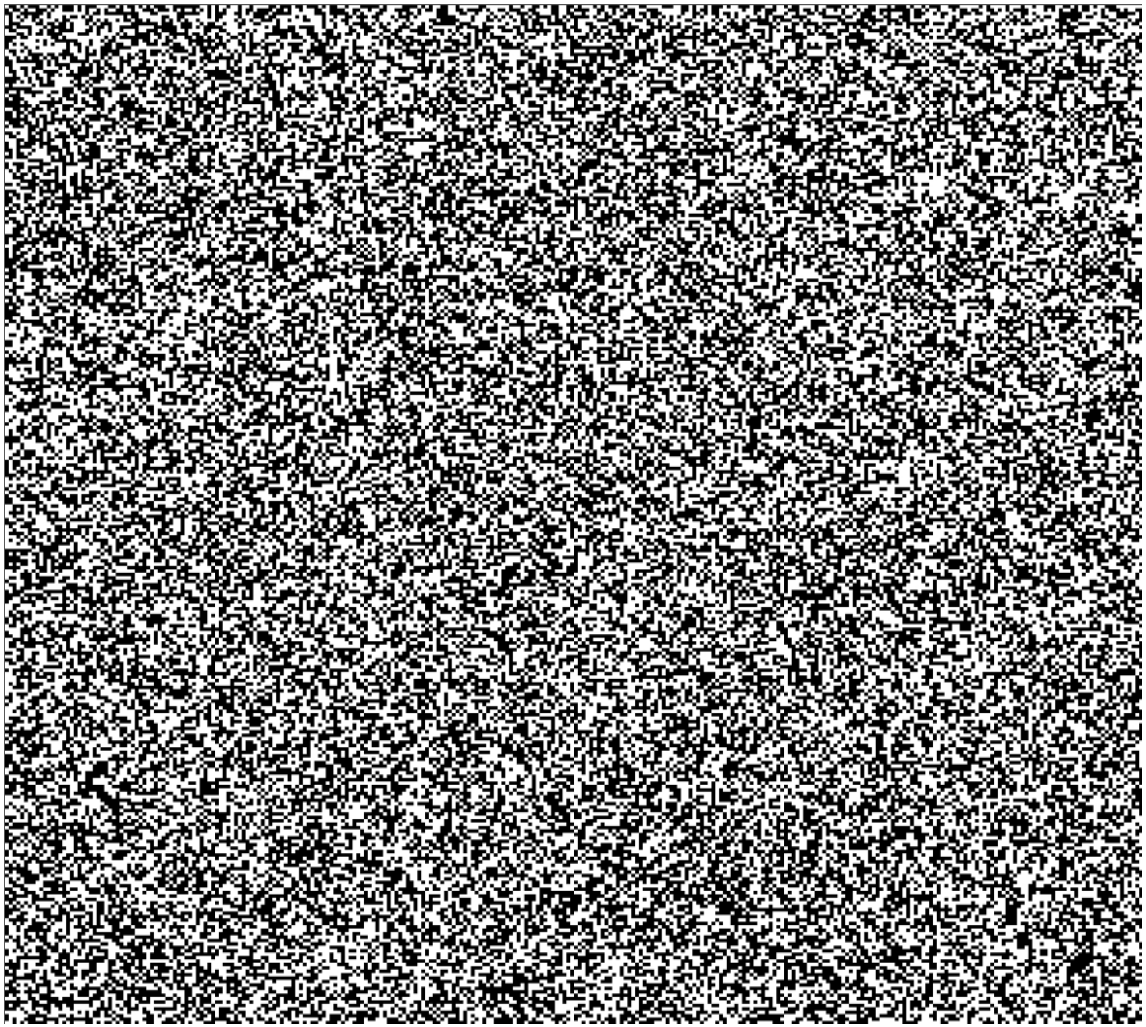


- Each pixel in this image comprises an 8-bit green colour component
- Examine the visual info stored in each bit to see if we can hide things

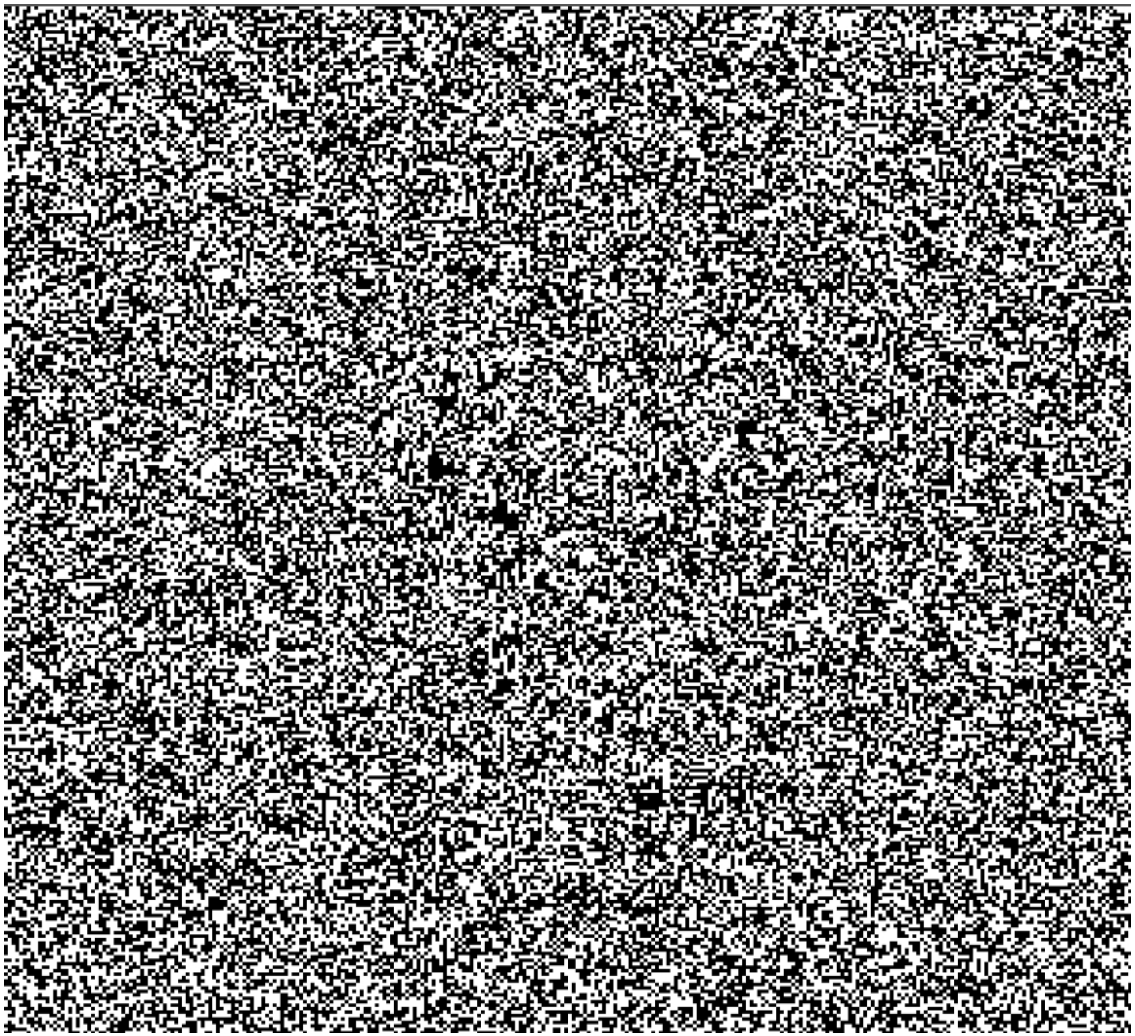
Bit 0 (Least Significant Bit)



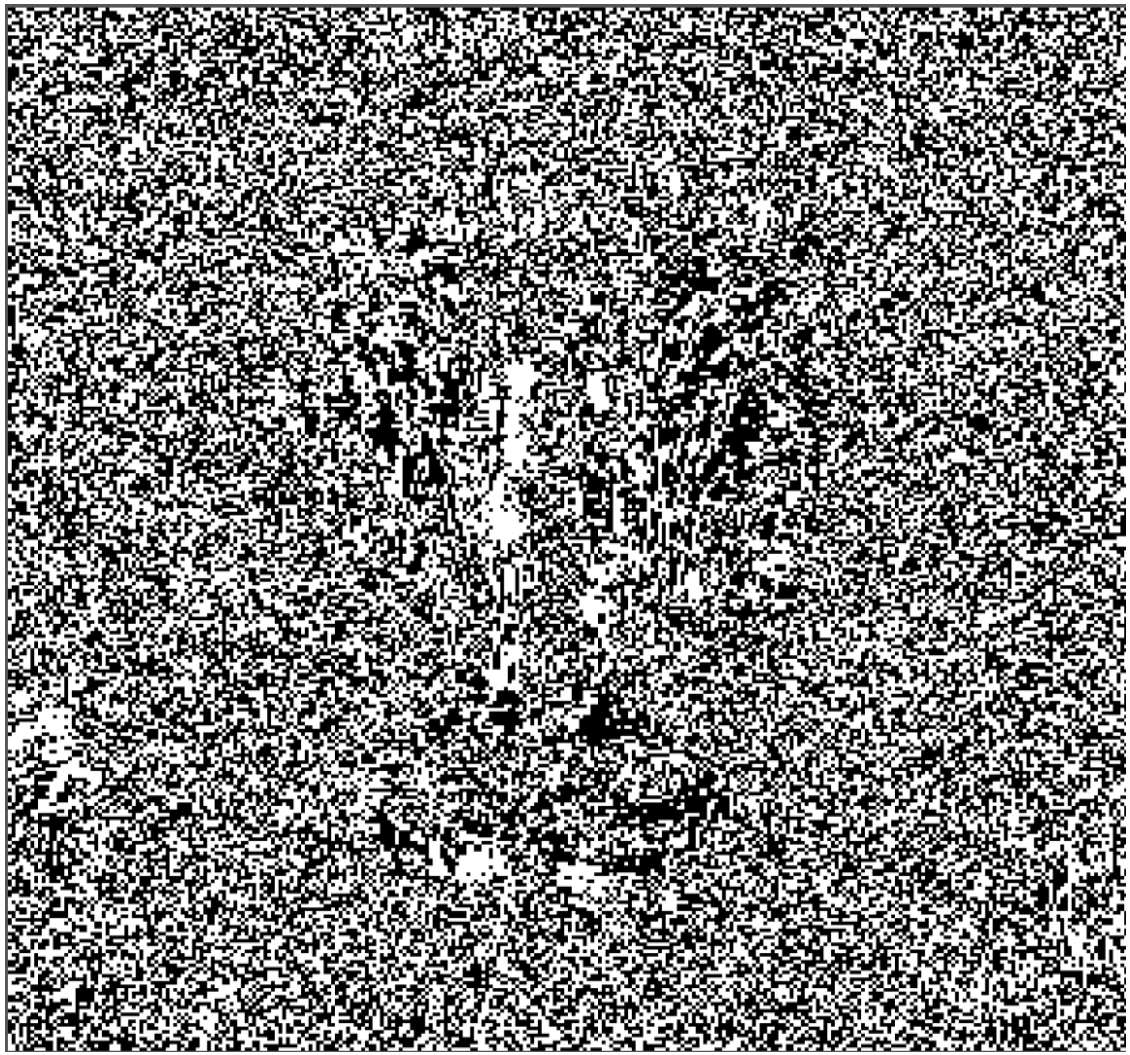
Bit 1 (of every pixel in the plane)



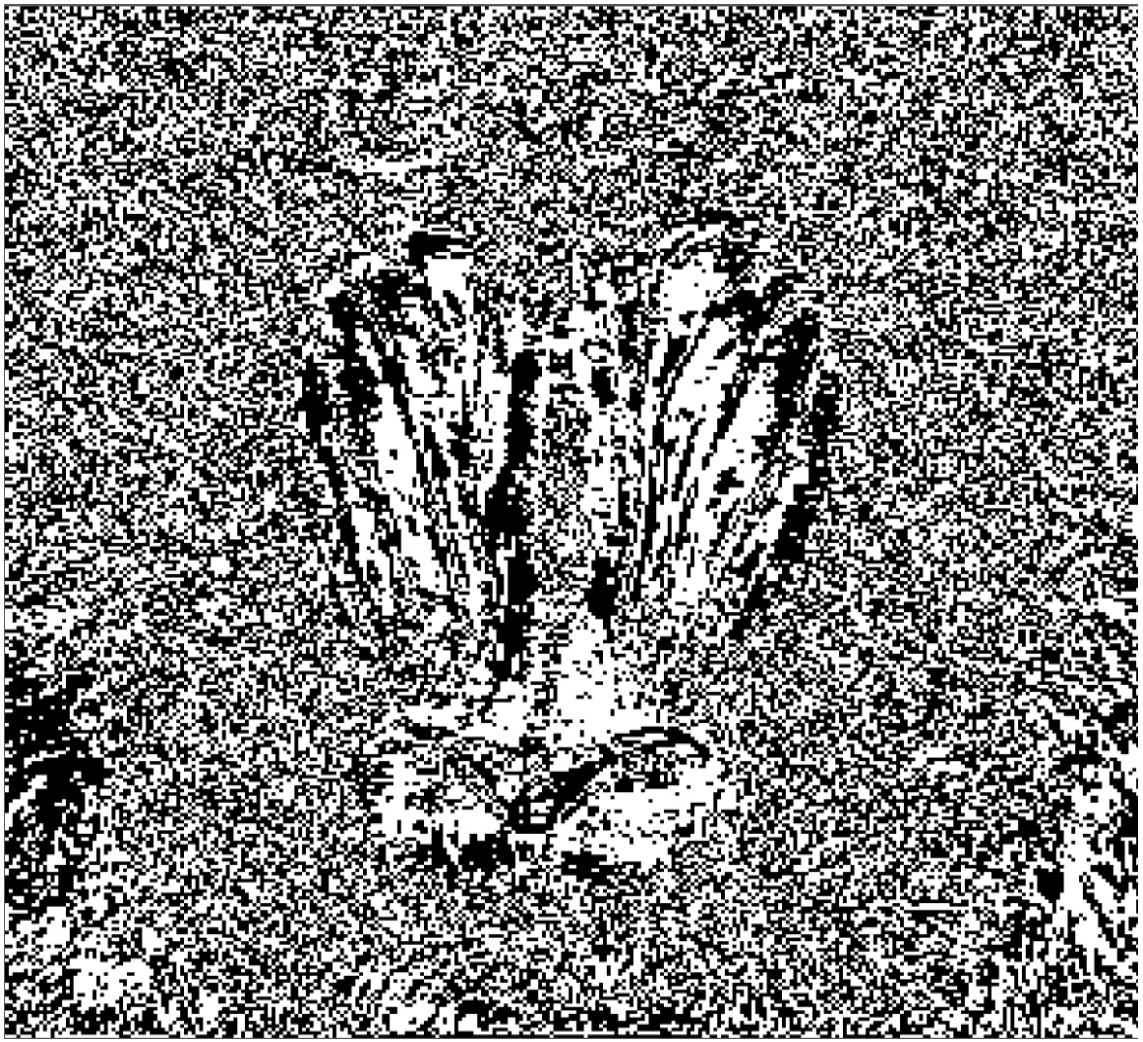
Bit 2



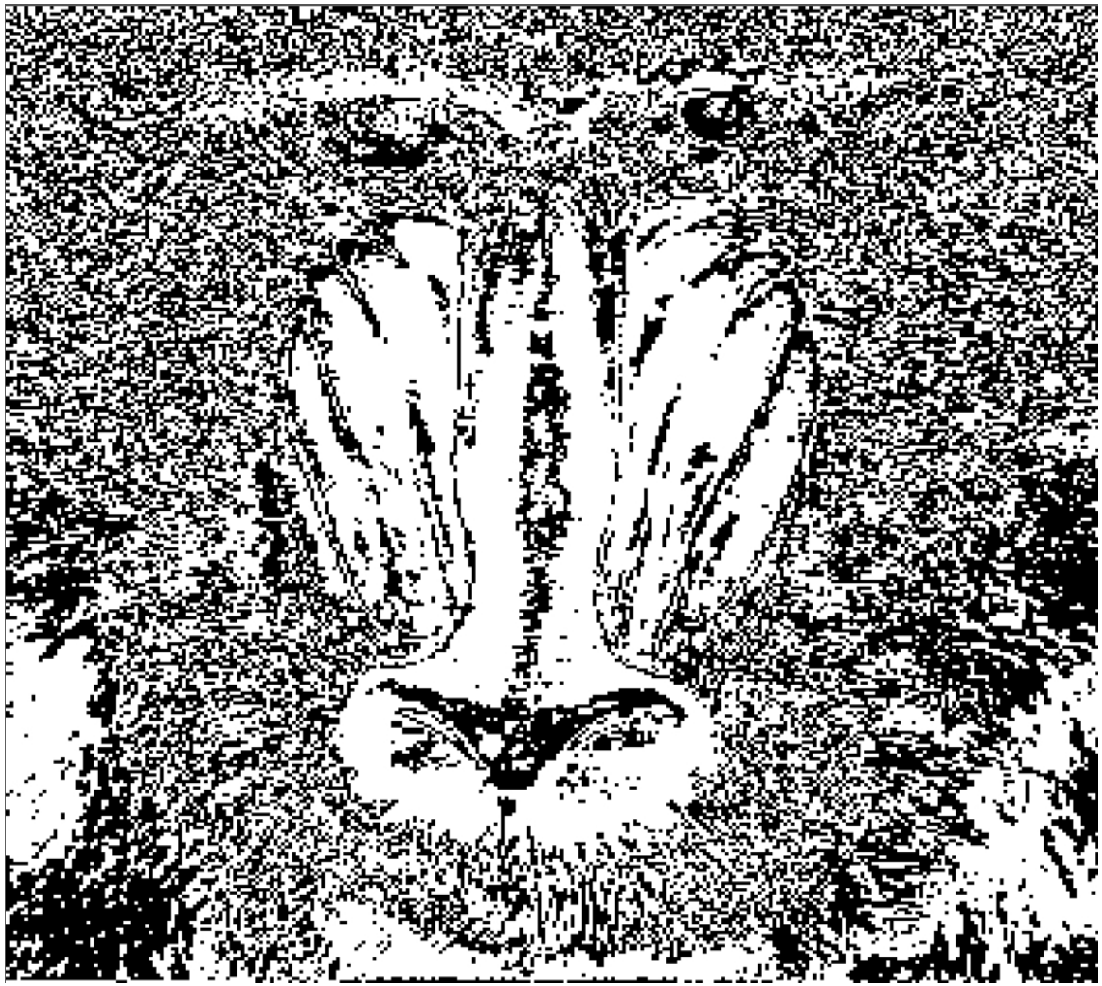
Bit 3



Bit 4



Bit 5



Bit 6



Bit 7

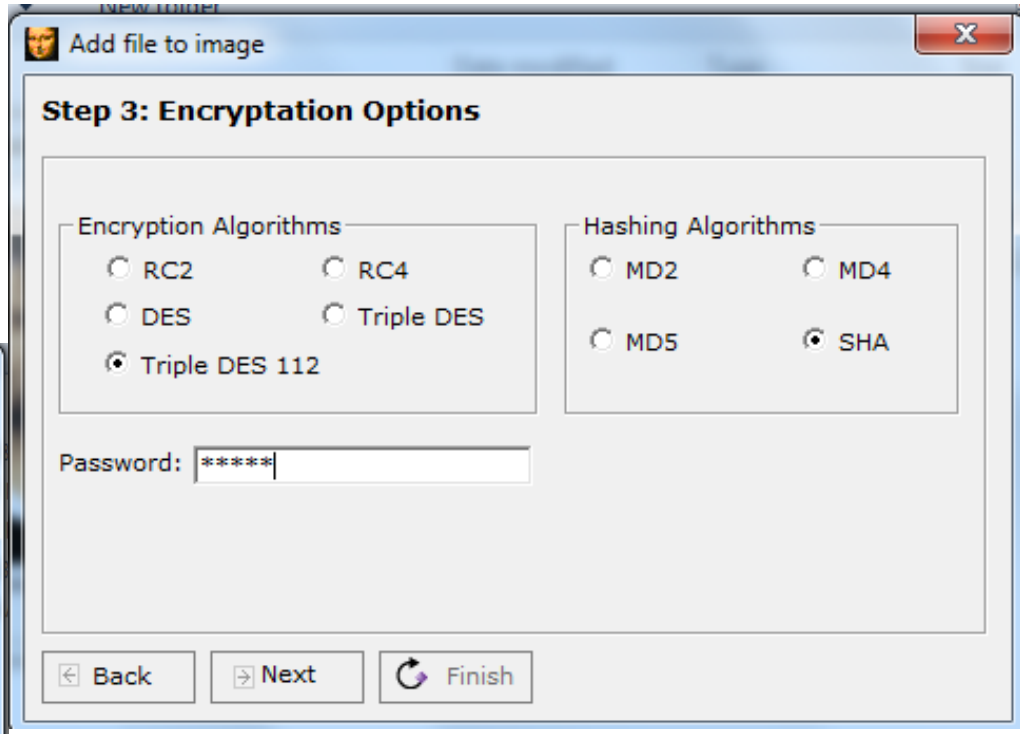


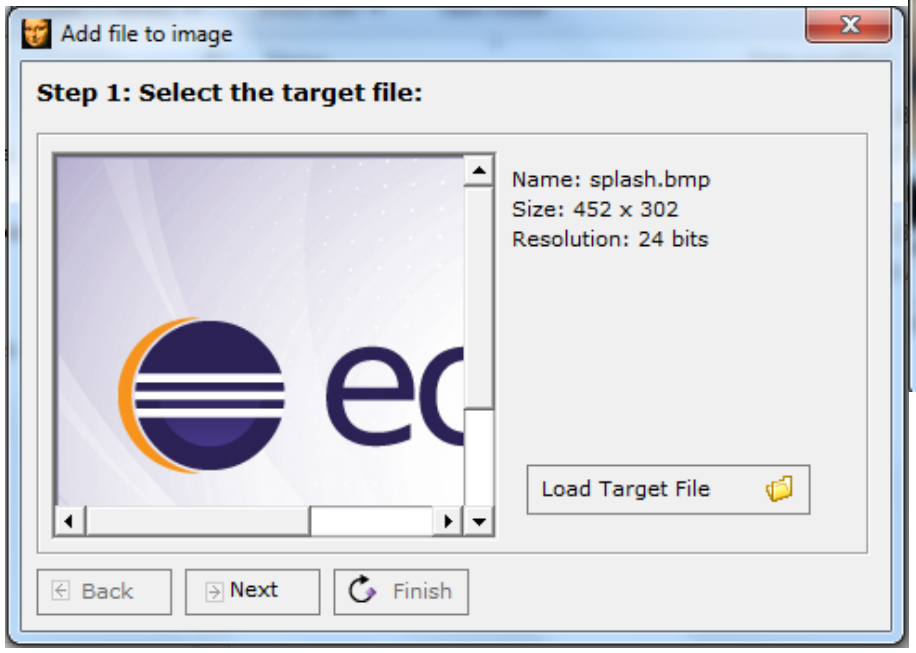
Similar colour info distribution for red and blue component planes

Make It Hidden – Xiao Steganography



http://download.cnet.com/Xiao-Steganography/3000-2092_4-10541494.html





- Hide system files, executables etc.
- Files are hidden in the image

Least Significant Bit (LSB)

- LSB of each colour component is often quite random
- Replacing it with a different value will likely not be noticed
- Replacing all the LSBs with payload bits is likely to result in an indifferent image (cover image)
  - Hence, payload storage capacity of 1/8 (12.5%)
- Replacing both bits 0 and 1 with payload bits will give a 25% storage capacity (2/8)

LSB Storage Issues

- The baboon cover image is quite good as the colours change quite rapidly over the whole image
- Some scenery-type pictures will have large areas where the colour doesn't change very much
  - Sky, sea, mountains
  - Pixel values do not change very much
- Replacing the LSBs with changing payload values will then be quite noticeable
- Public domain steganalysis tools are available

Steganalysis

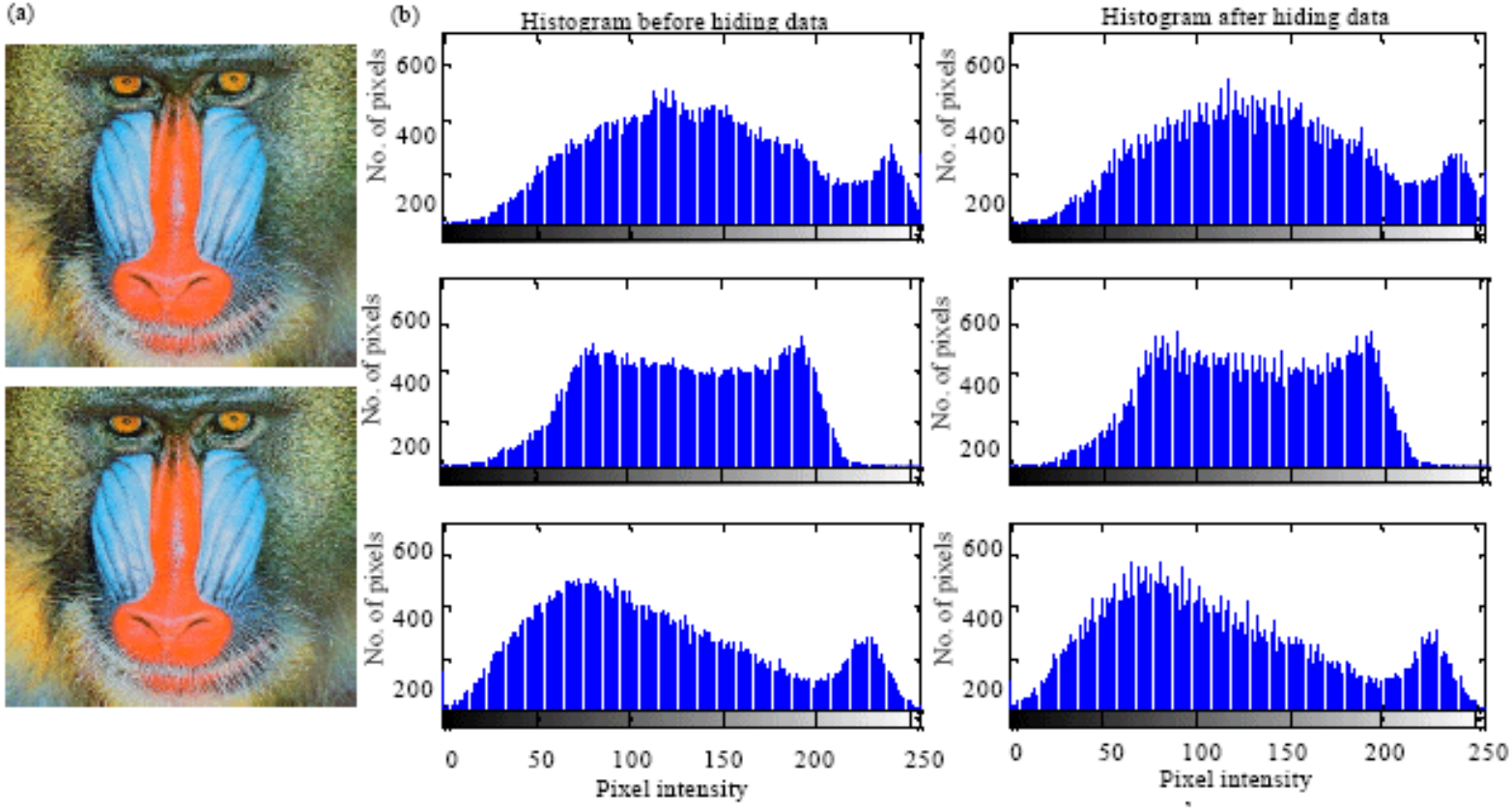
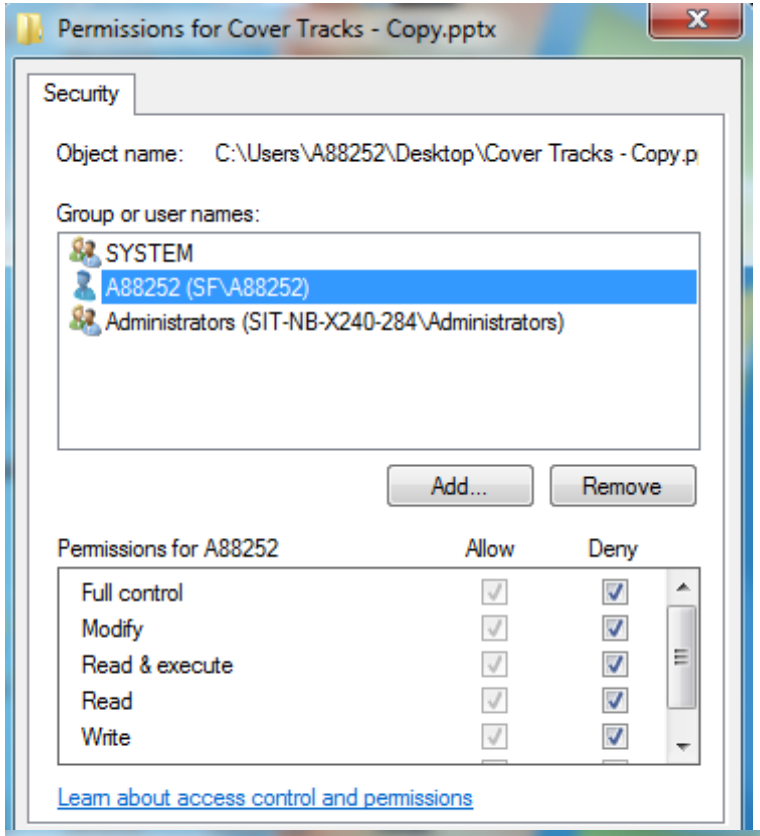
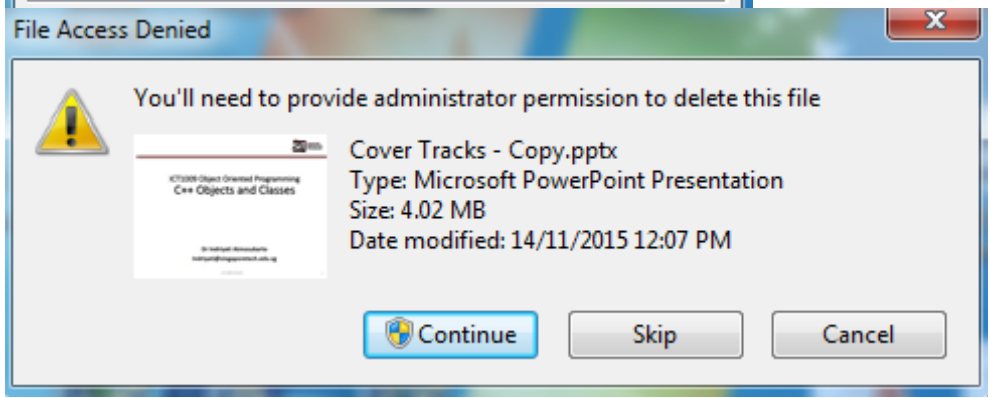


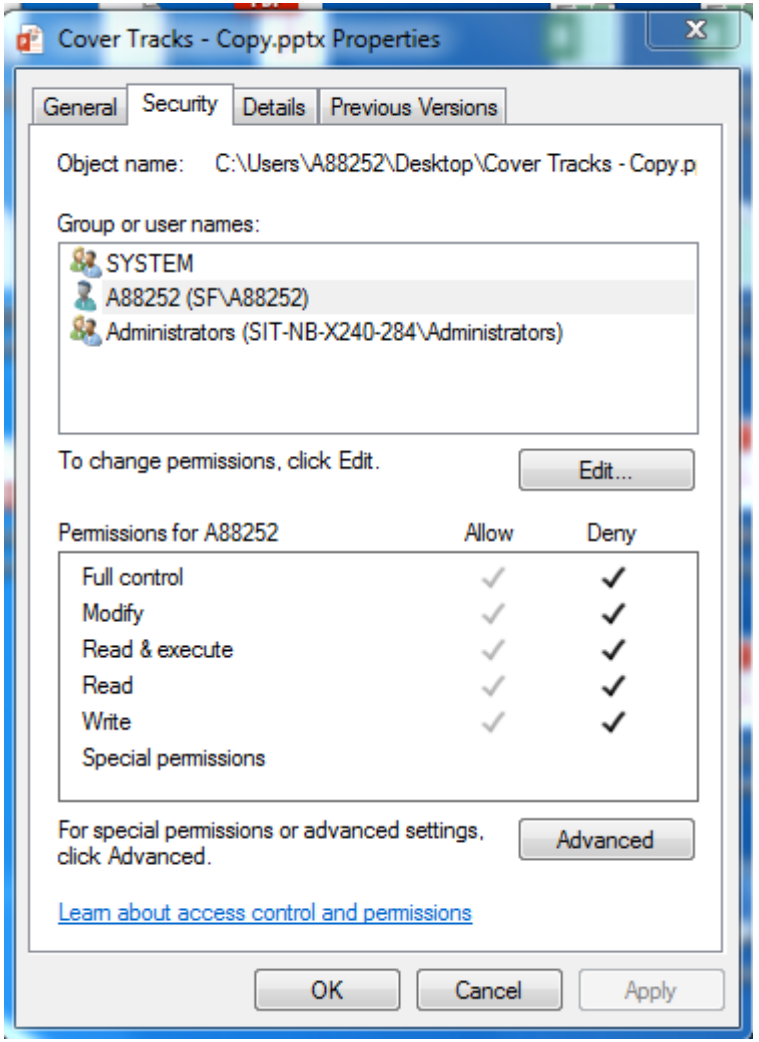
Fig. 8(a-b): (a) Baboon cover image and its corresponding stego image and (b) Subsequent histograms for cover and stego images

Undeletable Files and Folders

Make It Undeletable - Permissions



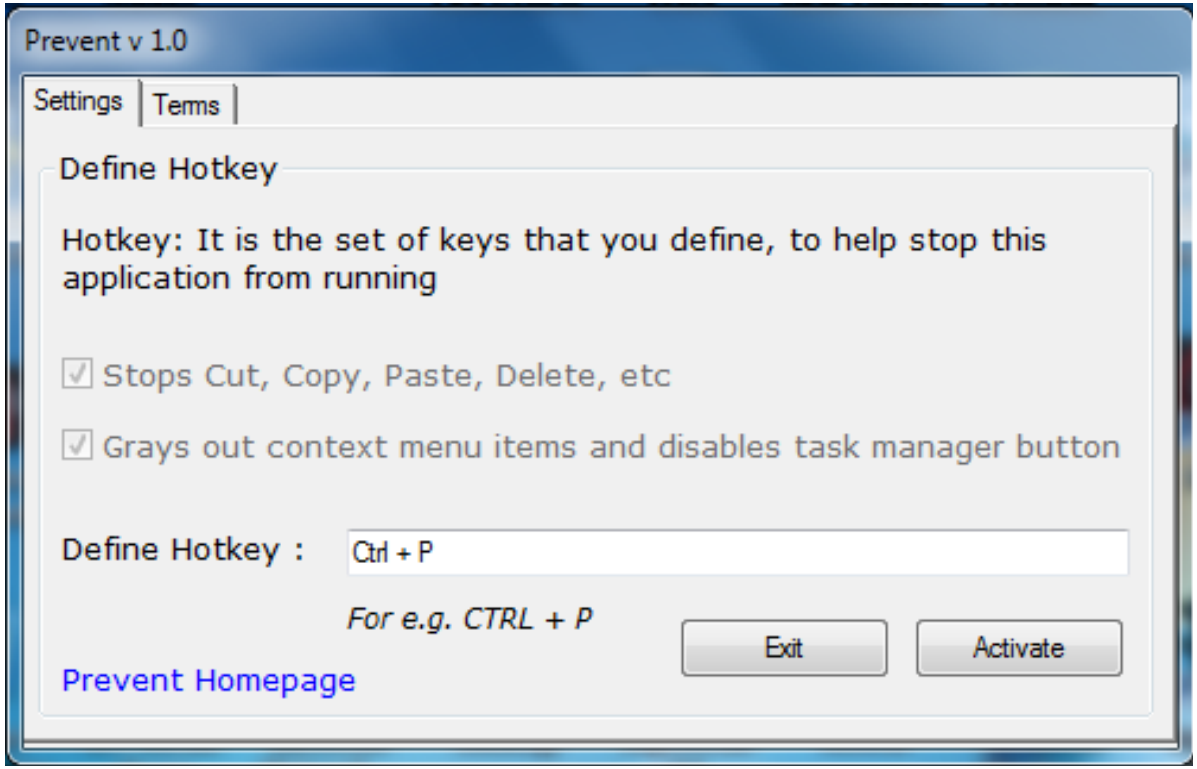




- Change permissions
- Rename and hide it

Make It Undeletable - Prevent

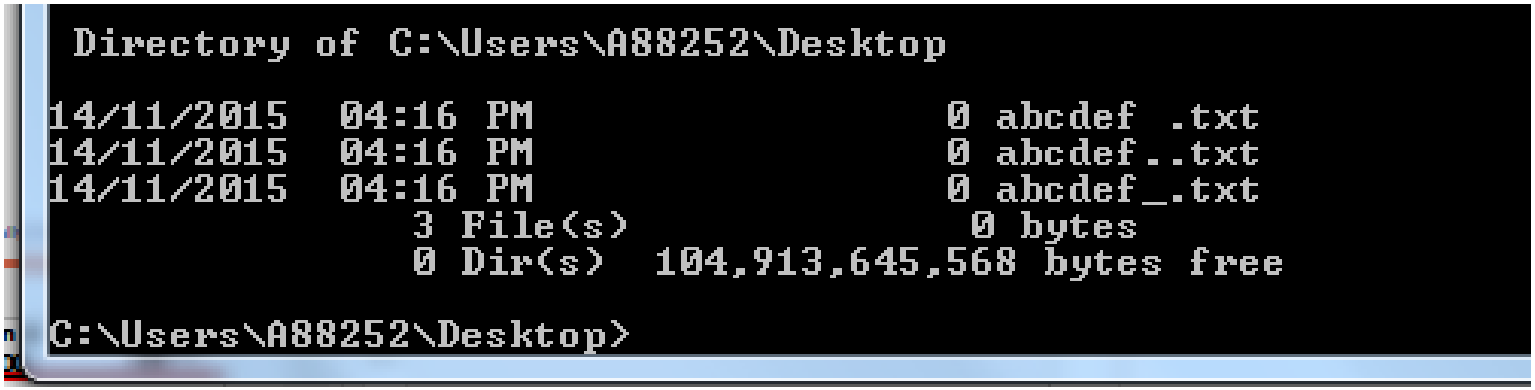
http://www.thewindowsclub.com/prevent-cut-paste-copy-delete-re-naming-of-files-folders



- Set hot key, e.g. Ctrl-P, to act as toggle
- Toggle prevent mode on and off to observe effects
- Essentially prevents activation of all functions via right mouse click

Make It “Undeletable” - Filename

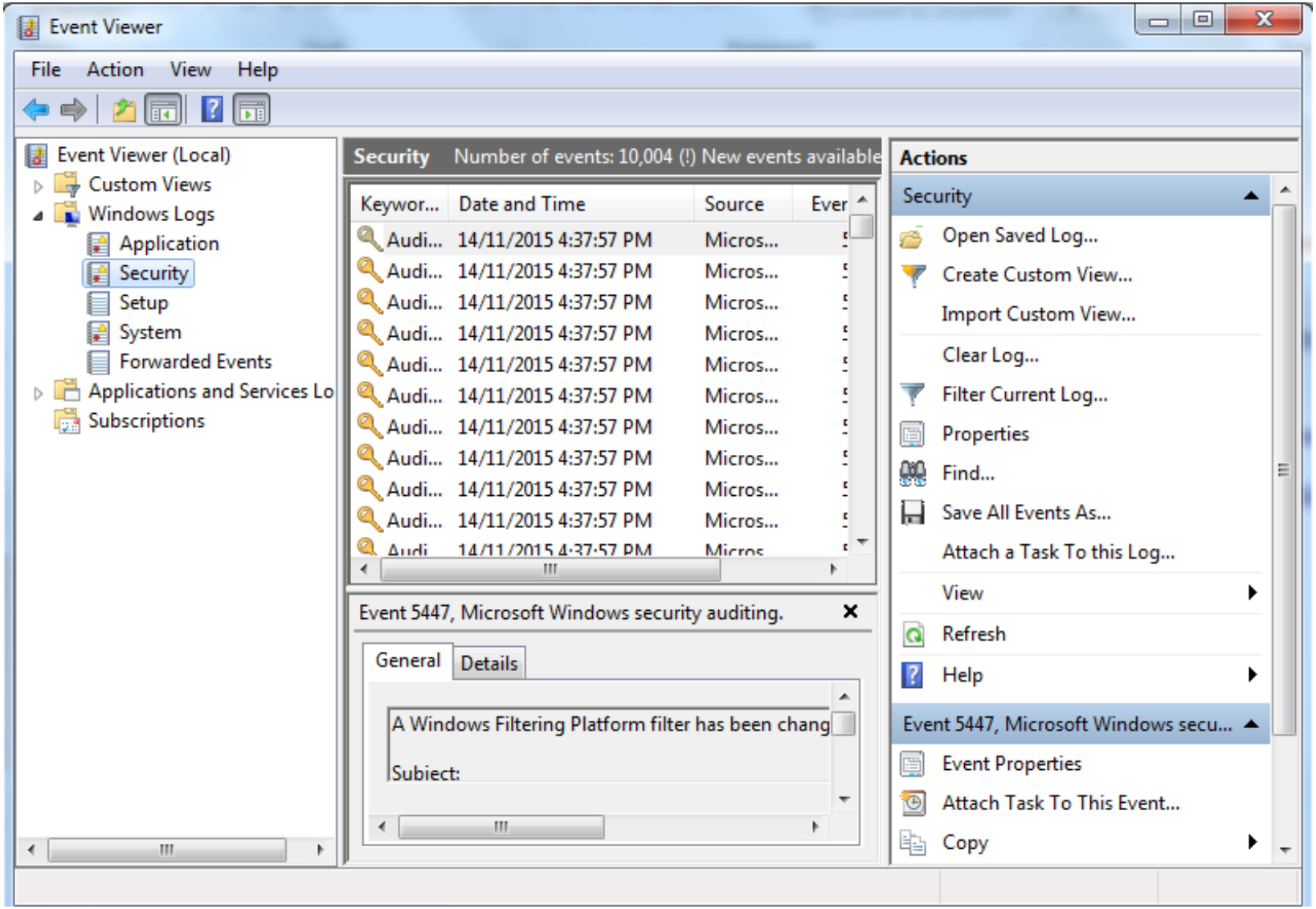
- Use file or folder name with inconspicuous characters e.g. space, period or underscore at the end
- Example filenames: abcde .txt, abcde..txt, abcde_.txt atc.
- C> dir -> What is displayed



- C> del abcdef.txt -> What is usually the initial attempt?

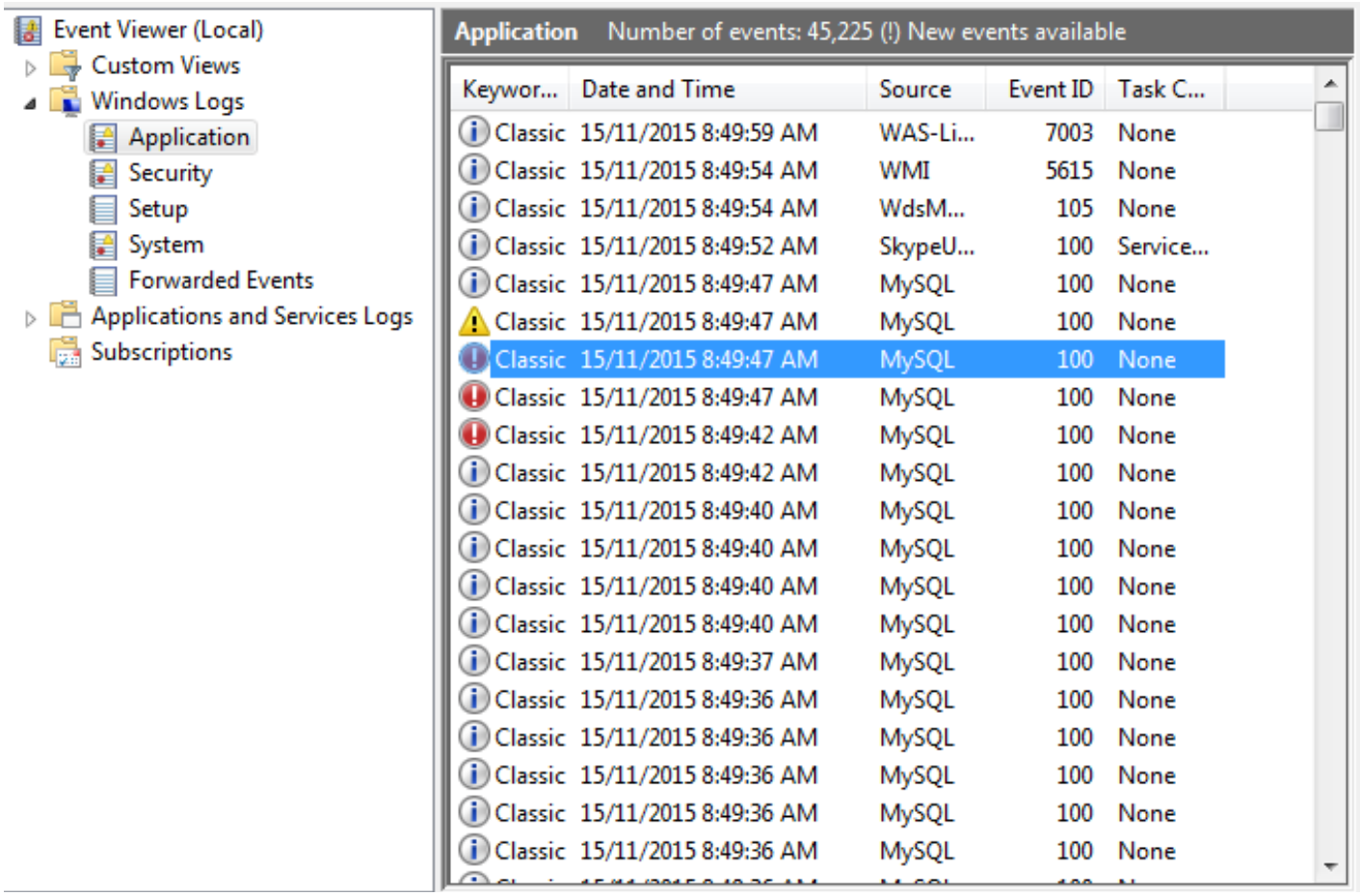
Log Files

Viewing Log Files



- Start -> Control Panel -> System and Security -> Administrative Tools -> Event Viewer

Differentiating Log Files



- Logs for application, security-related, setup and forwarded events are available
- Event type can be information , warning , error or audit

Log File Events

- Application events: classified as ***error***, ***warning***, or ***information***, depending on event severity. An error is a significant problem, such as loss of data. A warning is an event that isn't necessarily significant, but might indicate a possible future problem. An information describes the successful operation of a program, driver, or service.

- Security events: these are called ***audits*** and are described as successful or failed, such as whether a user trying to log on to Windows was successful.

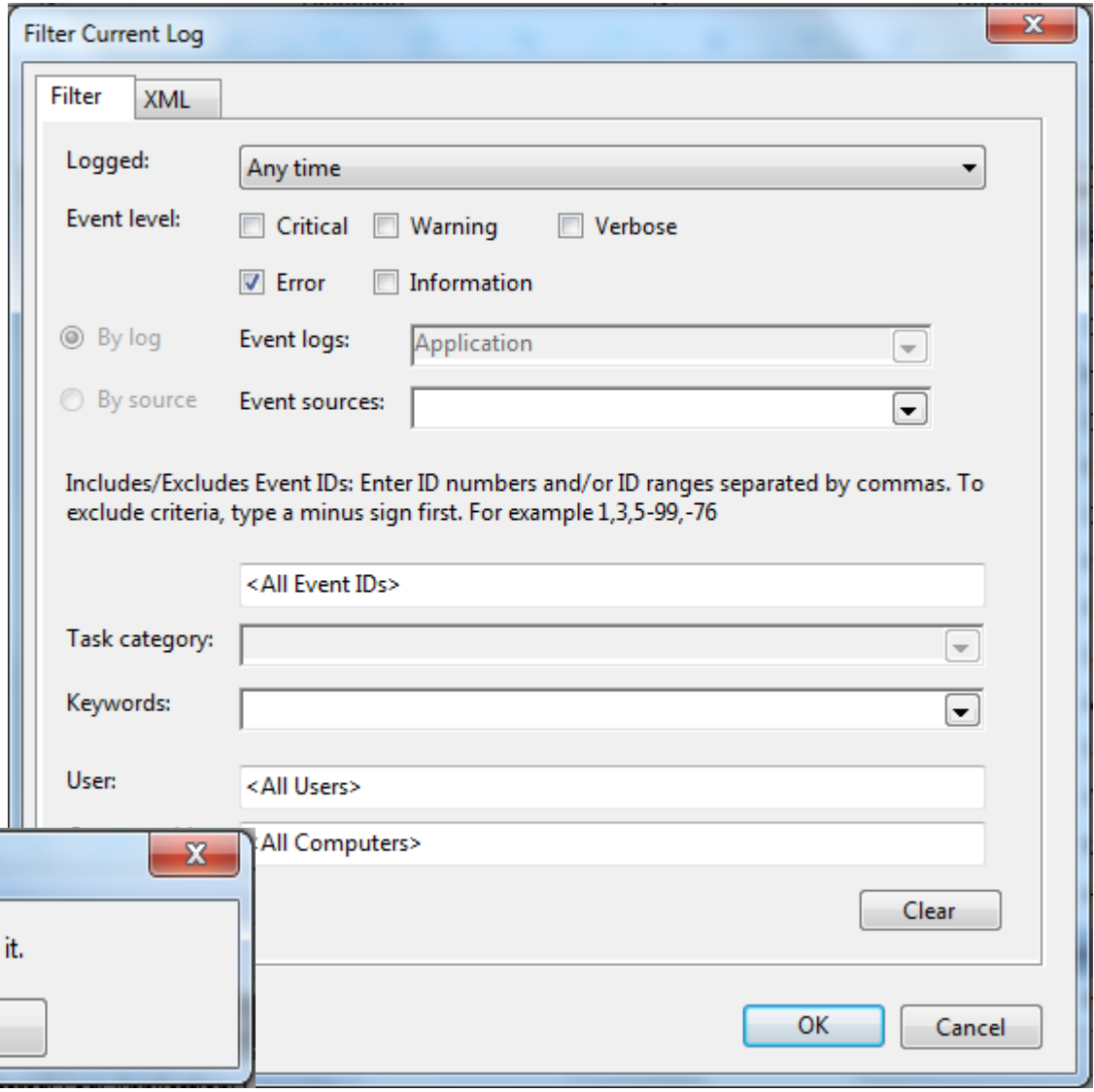
- Setup events: Computers that are configured as domain controllers will have additional logs displayed here.

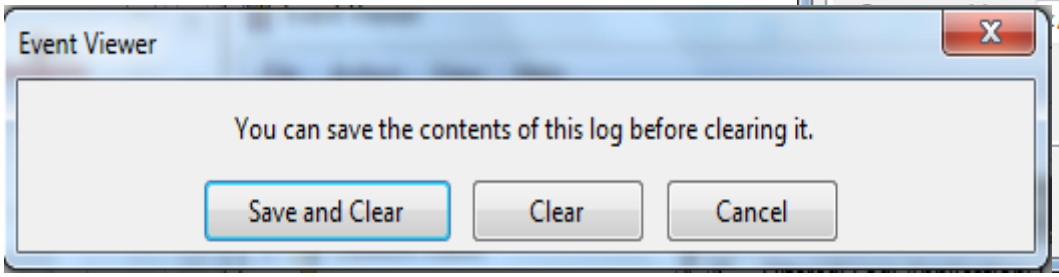
- System events: these are logged by Windows and Windows system services, and are classified as error, warning, or information.

- Forwarded events: these are forwarded to this log by other computers.

Filtering and Clearing Log Files

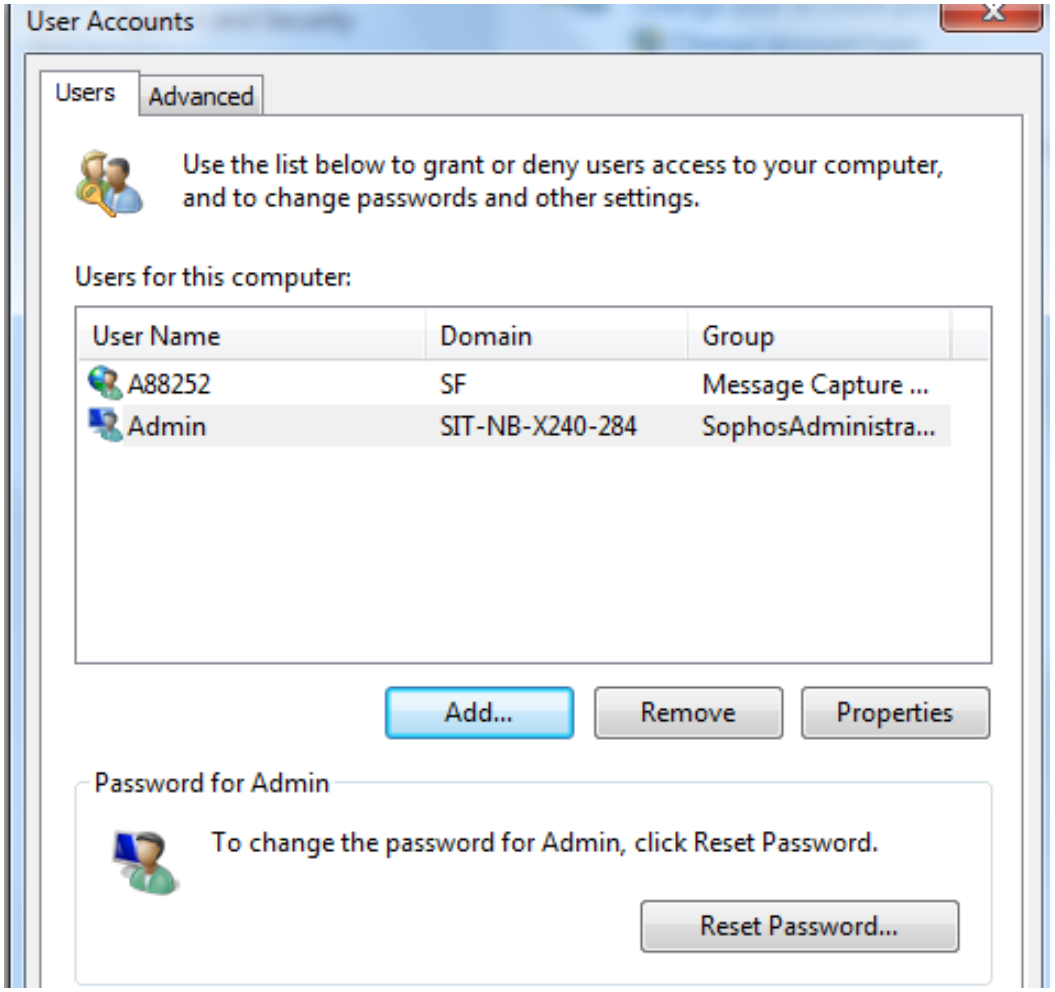
- Select the event level(s) log entries to filter for
- Click Action column entry on right, **Clear Log …**, to clear log file
- Opening or editing event log files cannot be done with a standard file editing tool
- http://ihackers.co/winzapper-covering-track-tools/



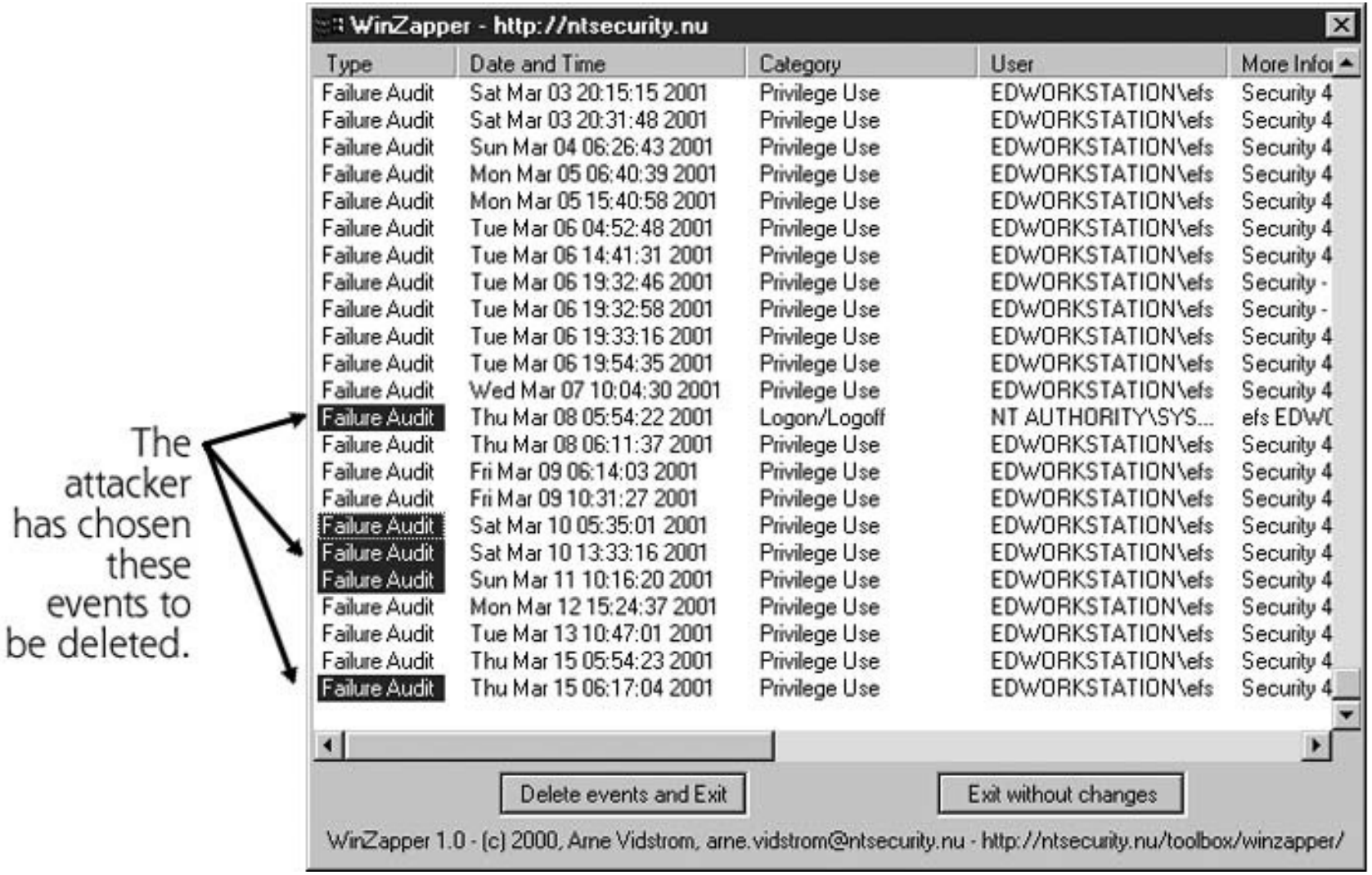


Editing Log Files – Winzapper

- Select the event level(s) log entries to filter for
- Click Action column entry on right, **Clear Log …**, to clear log file
- Opening or editing event log files cannot be done with a standard file editing tool
- http://getdntm.com/fwtr1?k=Winzapper+tool+download
- Need Admin account to run Winzapper – allows selective event deletion



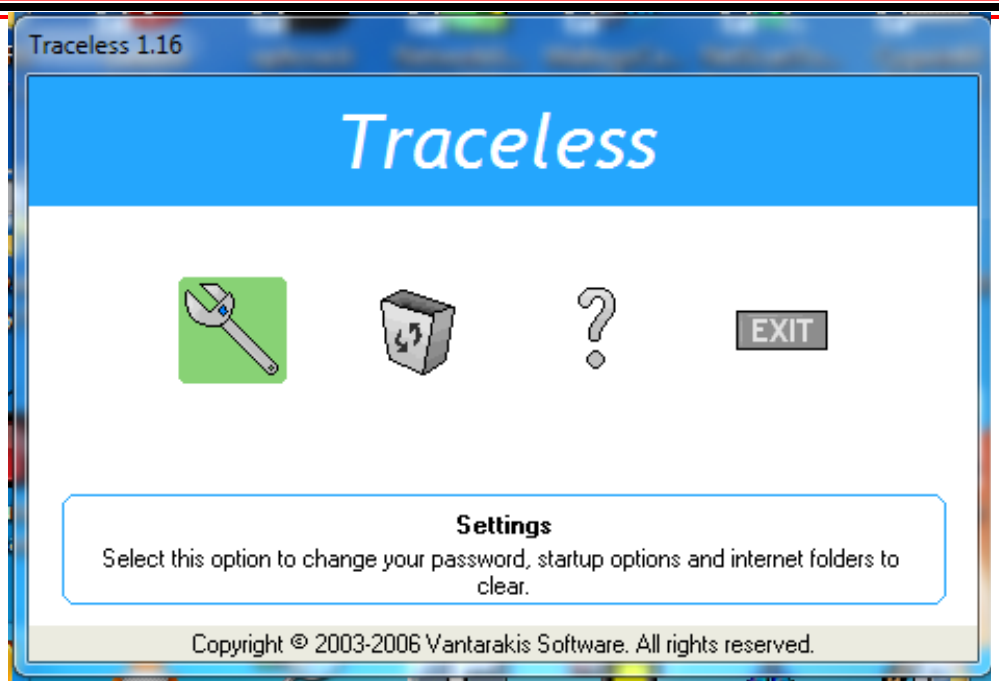
Winzapper – Selective Event Delete

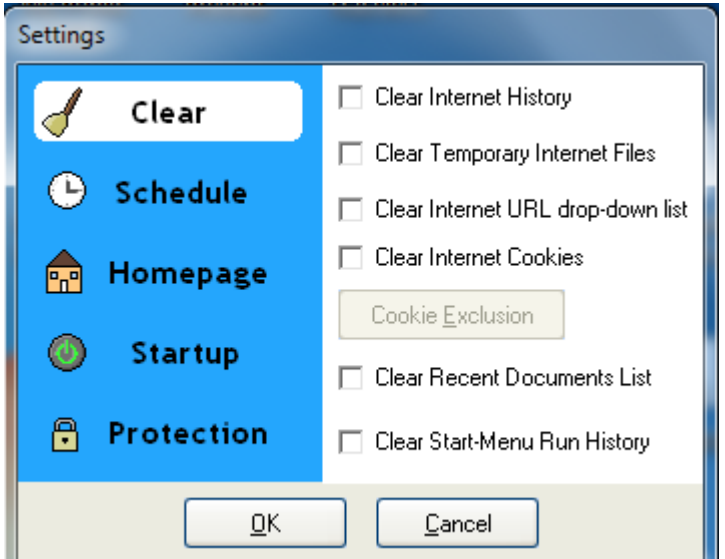


WinZapper tool lets an attacker selectively delete events from logs

Cleaning Up Target – Traceless

- Clear History
- Clear temporary internet files
- Clear cookies
- Clear Recent Documents list
- Password protected
- Run at Start-up
- Schedule to run when you want
- Comprehensive help file
- http://traceless.en.softonic.com/





Summary

- Hiding Files and Folders
  - DOS command attrib +h
  - Free Hide Folder and My Lockbox (folders only)
  - QuickStego and Xiao Steganography (files)
- Undeletable Files and Folders
  - Changing permissions
  - Prevent – disables right mouse functions
  - Add less visually distinct characters to filename
- Log Files
  - View and clear all
  - Winzapper – Selectively deletion but needs admin account

- **Next Lecture – APTs**